



SIMATS
ENGINEERING



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Sreevatha Institute of Medical And Technical Sciences
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Force-Torque Sensor

MLA0103-ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

GUIDED BY:
DR.PRAKASH MATHIALAGAN

PRESENTED BY:
T BHUPAL(19242523)

INTRODUCTION

- A Force-Torque (FT) sensor is a precision measuring device that acts as a "sense of touch" for automated machinery and robotics.
- It is designed to measure mechanical forces (push/pull) and rotational torques (twist) applied to a specific point.
- The most advanced and common type is a 6-axis FT sensor, which captures data across six degrees of freedom simultaneously.
- By translating physical contact into digital data, it allows robots to feel resistance and adjust their movements dynamically rather than moving blindly.



FORCE-TORQUE SENSOR IMAGE & COMPONENTS

- **Outer Shell:** The strong metal cover that protects the delicate inside parts from dust and damage.
- **Mounting Plates (Flanges):** The top and bottom flat pieces. One connects to the robot arm, and the other connects to the tool (like a robot hand).
- **Flexure (Metal Body):** A special metal piece inside that is designed to bend just a tiny, invisible amount when pushed.
- **Strain Gauges:** Tiny electronic stickers glued to the metal body that stretch or squeeze when the metal bends.
- **Circuit Boards:** The tiny computer brain inside that turns the stretching of the stickers into digital numbers for the robot.





HOW THE COMPONENTS WORK



- When the robot's tool touches an object, the push or twist transfers into the sensor.
- This push makes the special metal body (the flexure) bend a microscopic amount.
- Because the tiny "strain gauge" stickers are glued to the metal, they stretch or squeeze as the metal bends.
- When these stickers stretch or squeeze, the electricity flowing through them changes.
- The circuit board reads this changing electricity and immediately tells the robot exactly how much force it is feeling.



APPLICATIONS



- Putting Parts Together:** Helping robots slide pegs into tight holes or snap small parts together without smashing them.
- Polishing and Sanding:** Letting a robot press a sanding machine against a car door with the exact right amount of pressure.
- Medical Surgery:** Allowing doctors to "feel" what the robot arms are touching during remote surgeries.
- Product Testing:** Pressing buttons on a phone or keyboard repeatedly to test exactly how much strength is needed to click them.